

Itziar Morales

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Data Engineering & AI student (B.S. Software Engineering, minor Data Science & AI, U-tad, exp. 2027; GPA 8.07/10) with hands-on experience in Python data pipelines, SQL/NoSQL databases, big data processing (Hadoop, Spark/Scala), machine learning, and interactive data visualisation. Bilingual EN/ES (CEFR C2). Currently a Data Analytics & Engineering Intern at MGA, modernising legacy business workflows into a centralised relational data platform — schema design, data-integrity engineering, and Power BI analytics.

Technical Skills

Languages	Python, SQL, Java, Go, Rust, JavaScript/Node.js, C/C++, Scala, Bash
Data & ML	pandas, NumPy, scikit-learn, TensorFlow/Keras, matplotlib, seaborn, Plotly, ipywidgets, Tableau, Power BI, NLP, TF-IDF, CNN, transfer learning, feature engineering
Big Data	Apache Hadoop, Apache Spark (Scala), MapReduce, distributed data processing
Databases	PostgreSQL, MongoDB, Redis, Apache Cassandra, ChromaDB (vector store), distributed query optimisation
Infra & DevOps	Docker, Kubernetes, AWS (basics), Linux/NixOS, Git/GitHub/GitLab
Other	REST API design, JWT auth, Jest (unit testing), Postman, LaTeX, Agile/Scrum basics

Selected Projects

- ML Classification Suite** — kaggle.com/itzi97 2025
Three end-to-end ML pipelines: alphanumeric character recognition (scikit-learn, data augmentation), spam detection (NLP, TF-IDF, feature engineering), and image classification (CNN, Keras, transfer learning).
- Python Data Visualisation — Fitness Analytics** — github.com/itzi97/DV_Python 2026
Time-series performance dashboard with pandas, matplotlib, seaborn, Plotly and ipywidgets; includes data cleaning, aggregation, interactive widgets and trend analysis.
- Tableau Dashboard — US Firearm Homicides** — github.com/itzi97/DV_Tableau 2026
Interactive regional and temporal visualisations built in Tableau from raw CSV data; end-to-end data cleaning and narrative storytelling.
- BildyApp — REST API Backend** — github.com/itzi97/wpil-bildyapp 2026
Production-structured Node.js/Express API managing clients, projects and certifications; JWT auth, role-based access control, Jest unit tests, Postman collection, Dockerised deployment.
- AD-ODM — Python Object-Document Mapper** — github.com/itzi97/AD-ODM 2025
Lightweight ODM for MongoDB and Redis (pymongo, redis-py) with custom serialisation; reduces boilerplate for NoSQL data access layers.
- Distributed Systems & Containerisation** — github.com/itzi97/DS_exercise1_fm 2025
C++ distributed services deployed with Docker and Kubernetes; covers concurrency, messaging protocols, and container orchestration.

Work Experience

- Data Analytics & Engineering Intern** — MGA Jul-Sep 2026 | Remote
Migrating legacy spreadsheet-based workflows to a centralised PostgreSQL + Java 21/Spring Boot platform. Designed normalised relational schemas as a single source of truth, built a transactional ledger with database-enforced integrity constraints, and engineered safe data-migration procedures that prevent corruption during historical backfills. Also developed Power BI reports and predictive models and validated designs with business stakeholders. Stack: Java 21/Spring Boot, Python, PostgreSQL, Power BI, Docker, GitLab.
- Private Tutor — Mathematics & Programming** — Self-employed 2020–2021
One-on-one tutoring in calculus, algebra and Python/programming fundamentals; designed custom curricula and tracked student progress.

Education

- B.S. Software Engineering**, minor Data Science & AI — U-tad, Madrid 2025–2027 (exp.)
GPA 8.07/10 (8.61/10 past 2 years) | 4 Honors distinctions
Relevant courses: Machine Learning I & II, Data Engineering Projects, Information Retrieval & Analysis, Advanced Databases, Distributed Systems, Data Visualisation, Computer Vision, AI, Data Processing.
- Language Studies — English** — Escuela Oficial de Idiomas, Madrid 2022–2024
Advanced English programme; preparation for CEFR C2 certification.
- B.S. Computational Mathematics & CS**, minor Data Science & AI — U-tad 2019–2022
Pivoted to Software Engineering to deepen applied and systems-level expertise.
- B.S. CS — Real-Time Interactive Simulation** — DigiPen Institute of Technology 2015–2017
Left to pursue a broader mathematical and data-focused curriculum in Spain.

Certifications & Languages

Language certs	IELTS 8.5 (2026) CEFR C2 English (2024) Cambridge CPE (2014) CEFR B1 French (2013)
Spoken languages	Spanish (Native) English (Bilingual, C2) French (Intermediate, B1)
Music	ABRSM Grade 8 Piano (2013) ABRSM Grade 8 Violoncello (2012)